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The relative incidence of positive and negative word of mouth: A multi-category study

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Abstract

In 15 studies, positive word of mouth (PWOM) is more common than negative word of mouth (NWOM) in every case. The incidence ratio averages 3 to 1. Categories with high levels of NWOM tend to have high levels of PWOM, and individuals who produce NWOM also tend to produce PWOM. In follow-up studies, using two alternative measures of WOM, we confirm that PWOM occurs approximately three times as often as NWOM. Further exploratory studies indicate that most PWOM is about a consumer's main brand; as a consequence, the incidence of PWOM in a category is distributed according to brand market share. We discuss possible explanations for these results and develop six propositions to guide further research on word of mouth.
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1. Introduction

People receive information from their physical surroundings, from the mass media and through their dealings with other persons. Information from other persons about goods and services may come from sales personnel or from other consumers. In marketing, word of mouth (WOM) is normally used to describe advice from other consumers. The interactivity, speed, and lack of commercial bias of WOM make it a very effective source of information about prospective consumer choices, particularly with regard to services for which pre-purchase experience may be limited. In consumer choice, WOM is often the dominant factor; for example, Keaveney (1995) observed that positive word of mouth (PWOM) was the main source of information when people found a new service supplier.

Since WOM guides much of our consumption, it is important that marketers understand recurrent patterns in the incidence of WOM. There are indications that marketers believe NWOM is more common than PWOM; for example, Naylor and Kleiser (2000) titled their work "Negative Versus Positive Word of Mouth: An Exception to the Rule" when they found more PWOM than NWOM. However, if a large majority of goods and service users are satisfied, it is likely that PWOM is more common than NWOM. We seek to test this judgment with evidence. We also investigate how the incidence of PWOM and NWOM differs by category, individual consumer and brand. In addition, we determine whether a few people give WOM often or many people give it infrequently and what proportions of PWOM and NWOM are about respondents' main brands. We also observe how the production of WOM relates to market share. A continuing problem in this field has been the reliance on single recall measures; we use three different measures (one of which is not based on recall) to establish our results.

In summary, we seek answers to the following questions:

- What is the relative occurrence of PWOM and NWOM?
- Do categories that elicit more NWOM also elicit more PWOM?
- How does the production of PWOM and NWOM depend on the *penetration* (the proportion of the sample that produces it) and the *frequency* (the average production rate of those that produce it)?
- How is the production of PWOM and NWOM related to a brand's market share?

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- Is PWOM mostly about the consumer's main brand and NWOM mostly about other brands?
- Do individuals who produce more NWOM about a category also produce more PWOM about the category?
- Are similar results obtained when alternative measures of WOM are used?

We measure the *incidence* (occurrences in a period) of PWOM and NWOM. We investigate *users* of brands and choose *established* categories that are generally well known and used by most consumers. Thus, we are not concerned about the way in which WOM may be involved in the adoption of new categories, as studied, for example, by Rogers (2003).

There are a number of potential applications of this work. First, measures of the incidence of PWOM and NWOM may be used, together with measures of publicity, advertising, distribution, and other determinants of sales, to predict changes in demand for brands in a category. Second, evidence about the pattern of PWOM and NWOM may help managers promote PWOM and restrict NWOM and target segments of the user base that are particularly influential. Third, Bayus (1985) has shown that advertising may stimulate WOM, which suggests that marketers might be able to engender more WOM and raise the effectiveness of marketing communications if they had a better understanding of how WOM occurs. Fourth, we need to understand the general patterns evident in WOM in order to provide a basis for further theory development and research.

The paper is organized in two phases. In Phase 1, we first examine existing evidence on the occurrence of PWOM and NWOM and the circumstances in which PWOM and NWOM occur. Second, we discuss the expected linkage between the penetration and the frequency of WOM. Third, we describe measures and surveys and report results for 15 studies.

In Phase 2, we address issues arising from Phase 1 using further studies. First, we consider the possibility that the relative rates of incidence that we find for PWOM and NWOM could be distorted by a method artifact and report corroborative research using two alternative methods. Second, we report evidence on whether the brand cited in PWOM and NWOM is more often the respondent's main brand or another brand. Third, we present findings on how the incidence of PWOM and NWOM relates to brand market share.

Finally, we summarize our findings and discuss possible reasons for the patterns we find in WOM production. This summary leads us to develop six propositions to guide future research. We discuss limitations of our work and outline the implications of our observations for managerial practice.

2. Phase 1: review of related research

2.1. Positive information is more common than negative information

When markets are competitive, products do not easily survive if they cause dissatisfaction. As a result, unsatisfactory products disappear and most product experience is positive. Mittal and Lassar (1998) find that satisfied customers outnumber dissatisfied customers by more than four to one in the health care and car repair services. Peterson and Wilson (1992) conducted a comprehensive review of studies of satisfaction in the United States; in one table, they show that, on average, 83% of customers were satisfied, with the remaining 17% being distributed between neutrality and dissatisfaction. Thus, unless dissatisfied customers are exceptionally vociferous, we would expect PWOM to exceed NWOM because there are so many more satisfied customers.

We note two studies comparing the volume of positive and negative advice. First, Godes and Mayzlin (2004) studied comment on television shows on Web sites; they find that positive film appraisals occurred nearly twice as often as negative appraisals. However, comments on Web sites may not reflect common WOM. Second, Naylor and Kleiser (2000) studied users of a health and fitness resort. They find that, of the 97 respondents, 94 engaged in positive comments and 62 in negative comments. All those who gave negative comments also gave some positive comments. Among those primarily engaging in negative comments, the frequency of comments was half the level of those who just engaged in positive comments. However, the Naylor and Kleiser study shows a very high satisfaction level with the resort, and other resorts giving less satisfaction could therefore receive more NWOM. A more representative method of studying the relative incidence of PWOM and NWOM covers all the brands in a category.

2.2. Do dissatisfied customers advise more people than satisfied customers?

Marketing textbooks such as those by Heskett, Sasser, and Schlesinger (1997) and Hanna and Wosniak (2001) and consultants such as Silverman (1997) have reported WOM ratios of two or three to one when comparing dissatisfied and satisfied customers. The origin of these reports appears to be studies by the Technical Assistance Research Program (TARP). A report from TARP suggests that NWOM from dissatisfied customers occurs, on average, about twice as frequently as PWOM from satisfied customers, though the ratio varies with the category (Goodman & Newman, 2003).

Academic research shows a rather different picture. Two single brand studies indicate less NWOM from dissatisfied customers than PWOM from satisfied customers (Engel, Kegerreis, & Blackwell, 1969; Holmes & Lett, 1977). However, these results may reflect the particular character of the brand studied. Anderson (1998), using the Swedish Customer Satisfaction Barometer and the American Customer Satisfaction Index, which cover many industries, finds greater WOM among those who were very dissatisfied compared with those who were very satisfied but noted:

"The widespread belief in a high degree of word of mouth by dissatisfied customers may be unwarranted. In fact, in a sizable proportion of cases, the difference between the two is probably not significant."

Anderson's research reveals a U-shaped frequency of WOM across the satisfaction–dissatisfaction range. In the middle of this range, where respondents are neither satisfied nor

dissatisfied, the frequency of WOM production (among those producing it at least once) is still approximately 80% of the level that occurs with high satisfaction or high dissatisfaction. This finding suggests that factors other than satisfaction affect the production of WOM. Mangold, Miller, and Brockway (1999) illustrate some of these factors. They asked respondents to recall one positive and one negative communication made about a service in the previous six months, and then describe the circumstances that gave rise to the WOM. The predominant circumstance (50%) was the *felt need* of the receiver, usually prompted by a request for information. The next most common circumstance (19%) was *coincidental communication* when the advice arose out of a conversation. Mangold et al. find that the communicator's satisfaction or dissatisfaction was a primary circumstance in only 9% of the cases, and the receiver's satisfaction or dissatisfaction was primary in only 3% of the cases.

The work of Mangold et al. (1999) suggests that we should be careful about assuming that satisfied customers always produce PWOM and dissatisfied customers always produce NWOM. Because people often take into account the needs of others, they may give advice that is contrary to their feelings because the brand will, or will not, suit the other person. In some cases, they may give both positive and negative advice about the same brand to different persons. For example, we may recommend a Thai restaurant to someone who likes spicy cooking but advise against the restaurant when dealing with a person with less adventurous tastes.

2.3. Incidence, penetration, and frequency

The incidence of WOM depends on the proportion of the sample expressing it (the penetration) and the mean number of times those who express it do so (the frequency). It is useful to know whether the incidence occurs because many people (high penetration) give it infrequently or because few people (low penetration) give it very frequently. Thus, PWOM incidence might differ from NWOM incidence because of penetration, frequency, or both.

Normally, penetration and frequency are positively associated because these measures are statistically related. This effect is often seen as an association between the penetration and the frequency of purchase of brands in a category; bigger brands have greater penetration and purchase frequency than smaller brands, an effect called *double jeopardy* (Ehrenberg, Goodhardt, & Barwise, 1990). The association between penetration and frequency may also be found when we compare WOM about different categories; if a large proportion of a sample talks about brands in a category, we might expect those who talk about the brands to do so more frequently.

A number of conditions can affect the relationship between penetration and frequency. In the case of WOM, it may be difficult to work out a realistic NWOM penetration because many people may be unable to advise negatively through lack of knowledge of poor brands. Thus, an anomalous relationship between penetration and frequency may indicate a lack of relevant knowledge among users.

3. Phase 1: research

3.1. Method

Data on WOM were collected from users of the 15 categories listed in Table 1 (column 1) over the period 2001–03. All surveys were single-wave and were conducted in the United Kingdom. Samples from the general public were used, except in two cases where we used university samples. Some surveys covered two categories and these are reported as two separate studies. Questionnaires were delivered mainly by hand to households, with return by personal collection or free mail, but the surveys on restaurants and opticians used supplementary email and face-to-face distribution. For the ISP (general public), we used an Internet survey, and for the elementary school research, data were gathered via the schools (children took the questionnaire to their parents and brought it back to the school for collection). When hand delivery to households was employed, several locations were used to reduce location effects. Response rates (shown in Table 1) were generally high, ranging from 32 to 63%, except for the school study (14%) and the Internet study (2%).

3.2. Measures

In all cases, the form of the question used for PWOM was "In the last six months, how many times have you recommended any X?" For NWOM, we used, "In the last six months, how many times have you advised against X?" In the surveys, X was a dentist, bank, credit card provider, etc. For eight of the surveys, we used an ordinal response format: 0, 1, 2, 3–4, 5–8, more than 8. From responses, we derived an approximate figure for the number of WOM occurrences using scale midpoints or, in the case of the last response, $1.5 \times n$. For example, if "3–4 times" was ticked, we treated this as 3.5, and "more than 8 times" was treated as 12. We were concerned that this method stopped extreme responses, so we allowed extreme responses in the subsequent seven studies by using an open response format: "Please write in (0, 1, 2, etc.)…." Less than half of 1% of responses were extreme (numbers greater than 10), and all responses were used.

Our measures of PWOM and NWOM depend on recall and will be affected by measurement bias. If the proportionate errors associated with PWOM and NWOM are the same, the ratio measure will be true, but we cannot assume so. Our approach is to present results based on the recall of WOM in this phase and address this problem in Phase 2.

3.3. Phase 1 findings

In Table 1, the category is given in column 1 and the distribution method and response rate in column 2. In column 3, we show the number of respondents using the category. Columns 4 and 5 show the penetration and frequency of PWOM; column 6 gives the product of penetration and frequency, which is the incidence of PWOM. The next three columns provide the corresponding penetration, frequency and incidence data for NWOM. Column 10 gives the ratio of

Table 1
The incidence of PWOM and NWOM in 15 categories

Category [a]	Method [b] (Response rate %)	N	Positive WOM in last 6 months			Negative WOM in last 6 months			Ratio I_P/I_N (from columns 6 and 9)	Correlation r_s frequencies of PWOM and NWOM
			Penetration %	Frequency	Incidence per 100 I_P	Penetration %	Frequency	Incidence per 100 I_N		
1	2	3	4	5	6	7	8	9	10	11
Car insurance	DP (58)	109	24	1.5	36	1	2.0	2	18.0	−.05
Elementary school	DS (14)	122	73	3.4	248	12	1.6	19	13.1	.11
Dry cleaning	DCP (52)	111	28	2.2	62	5	1.2	6	10.3	.07
Car (university sample)	DP (48)	114	68	3.5	235	10	2.4	23	10.2	.12
Car (general public)	DP (57)	205	48	2.6	123	11	1.4	15	8.2	.22 *
Dentist	DP (57)	208	44	1.7	75	9	1.3	12	6.3	.15*
ISP (university sample)	DP (48)	73	33	1.9	61	7	2.3	15	4.1	.26*
Restaurant	DP+EF (63)	176	80	3.6	285	40	2.0	78	3.7	.59*
Car service	DCP (52)	131	37	2.3	87	10	2.4	24	3.6	.20*
Coffee shop	DP (43)	104	77	2.9	219	41	1.9	80	2.7	.45*
House cont. insurance	DP (58)	121	16	1.3	21	3	4.3	11	1.9	.07
Cell phone airtime	DP (45)	245	40	2.3	90	23	2.2	51	1.8	.49*
Optician	DP+EF (63)	87	36	1.5	55	21	1.5	31	1.8	.46*
Credit card	DP (43)	152	31	1.8	58	16	2.3	37	1.6	.59*
ISP (general public)	IS (2)	336	56	3.7	207	38	5.0	186	1.1	.18*
Means (unweighted)		153	46	2.4	124	17	2.3	40		0.26

* Significance <.05.
[a] Usually, two categories were investigated in the same survey (indicated by the same method and response rate). All samples are from the United Kingdom.
[b] Methods of gathering data: DP is drop off with post back; DS is distribution and collection via school; DCP is drop off and personal collection or post back; EF is email distribution plus face-to-face distribution and return; and IS is Internet study. The drop off and collect method is explained by Lovelock, Stiff, Cundiff, and Kaufman (1976).

PWOM to NWOM incidence, using the figures in columns 6 and 9. Column 11 shows the within-study correlations between the individual respondent frequencies of PWOM and NWOM. As a check on any effect of response rate, we correlated response rates with the ratio of PWOM to NWOM; the association was insignificant ($r=0.05$, $p=0.86$).

3.3.1. The relative incidence of PWOM and NWOM

From column 10, we see that PWOM incidence exceeds NWOM incidence in all 15 categories ($p<0.001$). We use the means at the bases of columns 6 and 9 to determine the ratio of PWOM to NWOM, that is, 124/40 or 3.1 to 1.[1] Thus we show that, for the 15 categories, consumers produce more PWOM than NWOM.

Because incidence is the product of penetration and frequency, we can ask whether the higher PWOM incidence occurs because of greater penetration, greater frequency, or both. In each category, the penetration of PWOM was greater than that of NWOM. The mean penetrations of PWOM and NWOM are substantially different, 46 versus 17% ($p<0.001$, two-tailed exact test), but the mean frequencies are nearly the same, 2.4 and 2.3 ($p=0.43$, two-tailed exact test). We conclude that there is more PWOM than NWOM because more people give PWOM rather than because those that give PWOM do so more frequently.

The lower penetrations for NWOM compared with PWOM could occur because many people do not know of negative examples that they could advise against; we return to this matter in the Discussion section.

3.3.2. The association between PWOM and NWOM

The association between the PWOM and NWOM incidences is significant ($r=0.53$, $p=0.04$), and therefore categories that generate more PWOM also generate more NWOM. This association occurs because the penetrations are significantly correlated ($r=0.68$, $p=0.005$); the correlation between frequencies is not significant ($r=0.15$, $p=0.58$).

It is possible for PWOM to come from one group of persons while another group gives NWOM. Alternatively, those individuals who give more NWOM may also give more PWOM. The correlation between PWOM and NWOM production at the individual level can be tested within each study. Column 11 shows that the associations are significantly and positively correlated in 10 of the 15 studies. We calculate that, across all studies, those who gave NWOM were 3.5 times more likely to be in the group of persons giving PWOM than in the slightly larger group of persons not giving PWOM. Therefore,

[1] According to a random effects meta-analysis of the 15 studies, the average proportions for PWOM and NWOM are 46 and 16, with standard errors of 5 and 3, respectively. The average frequencies are 2.3 and 2.2, with standard errors of 0.2 each. This result gives a PWOM to NWOM ratio of 3.0 to 1, which is close to the figure derived from the simpler analysis.

Table 2
Received and given PWOM and NWOM

Category (UK except as noted)	Method[a], (Response rate %)	Question type[b]	N	Incidence of WOM received in last 6 months			Incidence of WOM given in last 6 months		
				PWOM	NWOM	P/N	PWOM	NWOM	P/N
1	2	3	4	5	6	7	8	9	10
Restaurant	DP+EF, (63)	R/Aa	176	3.8	0.9	4.3	2.9	0.8	3.7
Luxury brands	CS, (87)	Ap/An	90	3.3	1.1	3.1	2.8	0.6	5.1
Holiday destination	DC, (83)	Ap/An	150	4.0	1.4	2.9	3.7	0.9	4.3
Restaurant	DP+DF, (23)	Af/Aa	95	2.3	0.8	2.9	2.8	1.0	2.8
Optician	DP+EF, (63)	R/Aa	87	0.6	0.2	2.6	0.6	0.3	1.8
Restaurant	DF, (53)	Af/Aa	157	2.5	1.2	2.1	4.7	1.1	4.2
Cell phone handset, China	DF, (75)	R/Aa	150	2.9	1.8	1.7	2.0	1.8	1.2
Bank	DF, (53)	Af/Aa	157	0.7	0.4	1.6	0.8	0.3	3.1
Coffee shop	DC, (83)	Ap/An	150	1.4	0.9	1.6	1.5	0.7	2.2
Cell phone airtime, China	DF, (73)	R/Aa	250	1.9	1.2	1.5	1.4	1.2	1.2
Car	DC, (83)	Ap/An	157	3.3	2.2	1.5	3.4	1.6	2.1
Supermarket	DP+DF, (23)	Af/Aa	95	1.1	0.8	1.4	1.3	0.7	1.8
Cell phone airtime	DC, (86)	Ap/An	166	4.0	4.0	1.0	2.6	2.8	1.0
Cell phone handset	DC, (86)	Ap/An	168	4.1	4.3	0.9	3.2	3.1	1.0
Credit card	DC, (86)	Ap/An	156	1.2	1.6	0.7	0.9	0.9	1.1
Means (unweighted)			147	2.5	1.5	2.0	2.3	1.2	2.4

[a] Methods of gathering data: DP is drop off with post back; EF is email distribution plus face-to-face distribution and return; CS is distribution and collection in a coffee shop; DC is drop off and personal collection; and DF is distribution via friends.
[b] Question types: R/Aa is recommend and advise against; Af/Aa is advise in favor and advise against; and Ap/An is advise positively and advise negatively.

those who advise against a brand are much more likely to recommend a brand as well.

3.3.3. The association between penetration and frequency

The correlation across categories between penetration and frequency for PWOM is positive and significant ($r=0.87$, $p<0.001$), but for NWOM, this correlation is not significant ($r=0.20$, $p=0.47$).

4. Phase 2 research

In Phase 2, we address issues arising from the research in Phase 1. We use alternative measures of WOM to test the finding that PWOM occurs more often than NWOM, and we examine the extent to which a respondent's main brand is the object of PWOM and NWOM. We also explore the incidence of PWOM and NWOM in relation to market share. In Phase 2, we also introduce some non-U.K. data and vary the phrasing of the questions about WOM to generalize our findings further. Data for Phase 2 were collected in 2004–05.

4.1. Alternative measures of word of mouth

Research on WOM relies on the ability of survey respondents to recall their past behavior; recall is prone to bias, and because the bias could differ between PWOM and NWOM, it might distort the measurement of relative incidence. There is no clear evidence that biases affect PWOM and NWOM differentially, but a number of mechanisms could produce this effect. Social desirability effects (Fisher, 1993) could favor recall of PWOM, and Rosen and Tesser (1972) have suggested that people avoid making negative statements for this reason. Also, Soderlund (1998) proposes that people process pleasant items more accurately and efficiently. Contrary to this argument, we note that receiving negative advice about brands is not normally unpleasant to the recipient. Also, the rarity of NWOM could give it surprise value or affect its diagnostic value, which might increase its recall. Aspects of an event that affect its retrieval from memory are discussed by Taylor (1982), and Godes and Mayzlin (2004) also review these issues.

To obtain unbiased measures, we should observe WOM as it occurs, but it is too infrequent in any category for this procedure to be feasible. An alternative is to induce WOM, for example in focus groups, but the moderation of focus groups could easily prime the production of one form of WOM at the expense of the other. Taking account of these problems, we selected two further methods for measuring the ratio of PWOM to NWOM.

4.1.1. Method 1: received WOM

Phase 1 results were based on WOM that was *given,* so an alternative approach surveys WOM *received*. Recall is involved in both giving and receiving, but the biases may differ between the two methods. In new studies, we asked respondents to report both received and given WOM in the last six months, and we used two different question phrasings about received WOM: *advise in favor* and *advise against*, and *advise positively* and *advise negatively.*

4.1.2. Method 2: conditional intention

In Phase 1, we found that the greater occurrence of PWOM depended on penetration, not frequency. One technique that offers a measure of penetration but not of frequency is to ask respondents to state what they *would do* under conditions that

Table 3
Penetrations of conditional WOM and given WOM

Category (UK except as noted)	Method[a], (Response rate %)	Question type[b]	N	Penetration of conditional WOM %			Penetration of WOM given in last 6 months %		
				PWOM	NWOM	P/N	PWOM	NWOM	P/N
1	2	3	4	5	6	7	8	9	10
Credit card, USA	DC, (78)	IA	125	89	8	11.0	100	13	7.7
Holiday destination, USA	DC, (78)	IA	128	95	13	7.3	91	22	4.1
Credit card	DC, (78)	IA	105	96	20	4.8	84	18	4.7
Holiday destination	DC, (78)	IA	104	82	24	3.4	86	24	3.6
Restaurant	DF, (53)	IS	157	64	29	2.2	86	37	2.3
Bank	DF, (53)	IS	157	21	12	1.7	29	9	3.2
Cell phone airtime	FF, (77)	IC	100	82	47	1.7	82	66	1.2
Restaurant	DF, (53)	IA	157	83	53	1.6	86	37	2.3
Cell phone handset	FF, (77)	IC	100	75	49	1.5	68	59	1.2
Bank	DF, (53)	IA	157	33	26	1.3	29	9	3.2
Means (unweighted)			129	72	28	3.7	74	29	3.4

[a] Methods of gathering data: DC is drop off and personal collection; DF is distribution via friends; and FF is face-to-face distribution.
[b] Question types: IA is "If asked, …"; IS is "If the subject came up, …"; and IC is "In a conversation, …".

often elicit WOM. An advantage of the conditional intention measure is that it does not require the recall of past behavior. Mangold et al. (1999) find that WOM most often arose from requests for advice or as an incidental outcome of a conversation. We can get close to these situations by using questions in the form: "If asked about X, would you recommend…?" and "If asked about X, would you advise against…?" Other questions used were "In a conversation, would you…?" and "If the subject of X came up in conversation, would you…?" We asked respondents to answer these conditional intention questions for both PWOM and NWOM as well as reporting on the PWOM and NWOM that they had given in the last six months.

4.2. Phase 2 findings

4.2.1. Comparing received and given advice

In Table 2, we report findings from 13 studies with U.K. respondents and two studies with Mainland Chinese respondents. Column 1 shows the category, column 2 the method of distribution and response rate, column 3 the question format used for asking about WOM, and column 4 the number of respondents.

Across categories, the incidences of received and given PWOM and received and given NWOM (columns 5 and 8, and 6 and 9) are significantly correlated ($r=0.77$, $p=0.001$; $r=0.97$, $p<0.001$, respectively). The correlation between the received and given PWOM to NWOM ratios (columns 7 and 10) is also significant ($r=0.73$, $p=0.002$). The ratios in columns 7 and 10 are not significantly related to the response rates of the different studies ($r=-0.22$, $p=0.44$, $r=-0.11$, $p=0.71$).

The ratio means (bases of columns 7 and 10) are 2.0 and 2.4. In two cases, the incidence of received NWOM is greater than the incidence of received PWOM, but overall, PWOM significantly exceeds NWOM for both received and given measures ($p=0.002$, $p=0.001$, respectively). We also find that PWOM exceeds NWOM for the two categories for which data were collected in China, which helps generalize our basic finding.

4.2.2. Comparing conditional and given intentions

We report data on conditional intentions from 10 studies. Table 3 lists the categories in column 1 and shows the method of distribution and response rate in column 2. Column 3 shows the question types used. Column 4 shows the number of respondents using the category. Columns 5, 6 and 7 show the penetrations of conditional PWOM and NWOM and their ratio, and columns 8, 9 and 10 show the corresponding figures for WOM given in the past six months.[2]

The PWOM to NWOM penetration ratios obtained by the two methods correlate significantly (columns 7 and 10, $r=0.90$, $p<0.002$). The averages of the ratio columns 7 and 10 are similar at 3.7 and 3.4. Consistent with our previous finding, the conditional intention penetrations for PWOM in column 5 significantly exceed the corresponding numbers for NWOM in column 6 ($p=0.002$, two-tailed exact test). The PWOM incidence exceeds NWOM incidence for studies conducted in the United States, thus further generalizing our findings.

4.3. Which brand is cited in PWOM and NWOM?

4.3.1. Findings

In 13 studies (which do not appear in a table), we asked whether the brand cited in the last instance of given PWOM was the respondent's main brand, then we repeated the question for NWOM. These studies covered cars, dentists, banks, cell phone handsets (twice), cell phone airtime (three times), cameras, computers, luxury leather goods, and restaurants (twice). On

[2] In the case of the bank and restaurant, we used two question types each to establish conditional intention, so the data for given WOM are the same for these rows in Table 3. Therefore, we reduced the data to eight rows by averaging before we tested for associations between the columns of data.

Table 4
Ownership of the brand cited in WOM

Ownership	Cell phone handsets		Cell phone airtime		Computers		Leather goods		Cameras		Means	
	PWOM %	NWOM %	PWOM %	NWOM %	PWOM %	NWOM %	PWOM %	NWOM %	PWOM %	NWOM %	PWOM %	NWOM %
Current brand	81	20	79	19	70	24	63	7	88	33	76	21
Previous brand	5	45	4	32	15	14	10	24	8	17	8	26
Earlier brand	10	26	5	24	4	5	19	45	4	17	8	23
Never owned	5	9	12	26	12	58	8	24	0	33	7	30

average, PWOM referred to the main brand in 78% of the cases, whereas NWOM was about a brand *other* than the main brand in 77% of the cases. Thus, as a rough approximation, four fifths of PWOM and one fifth of NWOM is about the main brand. In the three studies involving repertoire categories (restaurant [twice] and leather goods), the main brands were the object of PWOM somewhat less often (average 60%) than in subscription categories (average 83%); it seems likely that repertoire categories show this pattern because recommendations are shared across the brands in the repertoire.

In five cases, we had more detailed evidence about which brand was cited in WOM, and this evidence is shown in Table 4. On average, only 16% of PWOM is about previously owned brands, and 7% is about brands that have never been owned. By contrast, 49% of NWOM was about previously owned brands and 30% about brands that have never been owned.

If most PWOM is about the current main brand, PWOM incidence will reflect market share: big brands with more owners will generate more PWOM. In the case of NWOM, previously owned brands reflect the market shares that applied previously, and therefore, unless brand shares are rapidly changing, we would expect that the occurrence of NWOM would tend to follow market share, though the large proportion of NWOM about brands that have never been owned may disrupt this relationship. In Table 5, we explore the relationship between market share and WOM incidence in five categories, using the brands currently owned by the respondents to provide a market share measure. The WOM rates are the averages of WOM received and given. We see that in all cases, PWOM is closely related to market share, and this close relation occurs for three of the cases of NWOM. In two cases (cell phone airtime and cameras), there is no association between market share and NWOM, which suggests that particular problems had emerged for some brands in these categories. (We followed up this explanation in the case of airtime suppliers and found that two smaller brands, *Orange* and *3*, were criticized for service and coverage deficiencies.)

Table 5
Correlations (r) between market share and the rate of WOM

Category	PWOM	NWOM
Cell phones	0.97*	0.85*
Cell phone airtime	0.90*	0.13
Cameras	0.91*	0.02
Coffee shops	0.99*	0.97*
Computers	0.94*	0.87*

*$p<0.05$.

5. General discussion

5.1. The importance of multiple methods

Research findings that are based on a single category may relate to particular features of that category. To establish generalizations, we need to study a number of categories. Similarly, consistent results across a variety of population samples, using different methods of measurement, help support a generalization. Such generalizations show empirical patterns that can provide a basis for prediction as well as theory development. The value of such empirical patterns has been illustrated by different authors in a special issue of *Marketing Science* (1995, Vol. 14, Issue 3).

Our research covers a number of categories and many samples and employs different methods for measuring WOM and distributing questionnaires. Each study may have weaknesses – for example, the distribution of questionnaires via friends could produce some shared responses – but overall, the wide empirical basis gives greater weight to the evidence. We now suggest explanations for our findings, following Bass's (1995) "Empirical–Theoretical" route. We summarize our results and then discuss and develop six propositions for further research.

5.2. Summary of results

We find that PWOM is more common than NWOM. This occurs because there are more people producing PWOM rather than because those individuals who produce PWOM do so more frequently than those producing NWOM. We corroborate our basic finding that PWOM is more common than NWOM by using two other measures of WOM. Categories receiving high levels of PWOM tend to receive high levels of NWOM, and those individuals who give NWOM are much more likely to give PWOM. We observe that PWOM is usually about the current main brand; NWOM is usually about past brands or brands that have never been owned. The occurrence of PWOM, and sometimes NWOM, is positively related to market share.

5.3. Propositions for further research

In the past, research on WOM often has been associated with customer satisfaction and dissatisfaction. The picture that emerges from our work attributes more importance to the salience of the category in everyday life, as well as to the interest and knowledge that people have about the category. Satisfaction

is just one of several factors contributing to the production of WOM. We suggest that if a category is important to many people, advice will be more valuable, which raises the rate of both PWOM *and* NWOM. Similarly, if the brands in a category differ significantly, it will be more useful to recommend better brands *and* to warn others about weaker brands. If a category is frequently encountered, this should raise the incidence of both positive *and* negative advice.

Because those individuals who give NWOM are more likely to give PWOM, there are likely to be common individual-level determinants. Those who value and have frequent experience with a category, understand it well, and have more social contacts are likely to give more PWOM *and* more NWOM. These points are expressed in Propositions 1 and 2.

Proposition 1. *Production of WOM at an aggregate level depends on the relative importance of the category to consumers, the differences between brands within the category and the prominence of the category in the everyday environment.*

Proposition 2. *Production of WOM at an individual level depends on the value, knowledge and experience of the category, as well as on the range of the individual's social contacts.*

The association between the production of PWOM and market share is also important, though not surprising. Big brands have more customers, and customers usually recommend their main brand. Across five categories, the correlation between market share and PWOM is very high (Table 5). In the case of NWOM, the relationship with market share is less clear. Much of NWOM is about brands that have been used in the past, which would produce a positive association between NWOM and market share when brand shares do not change rapidly. But a large minority of NWOM is about brands that have never been owned, which will reduce any association between NWOM incidence and market share. Our data in Table 5 show that NWOM is correlated with market share in three out of five cases.

These data suggest that market share provides a benchmark for assessing comments about brands. Superior brand performance occurs when the rates of PWOM and NWOM run, respectively, above and below the market share rates. We propose Proposition 3 as a norm against which brands may be assessed.

Proposition 3. *Within a category, the incidence of PWOM and NWOM for each brand tends to be a function of that brand's market share.*

Although our findings confirm a positive association between penetration and frequency for PWOM, this relationship does not occur for NWOM. A possible explanation is that respondents often lack negative examples that could serve as the basis for NWOM. In some categories, such as cell phone handsets and airtime, most brands are familiar to users, so they may make negative judgments about these brands more easily. In these cases, the incidence of PWOM and NWOM may be closer (as they were in our work). In other categories, such as dentists, consumers are likely to lack the knowledge required for NWOM, and correspondingly, we find much more PWOM than NWOM about dentists. If many people lack experience of negative examples, NWOM penetration figures will be artificially low, because they will have been calculated using all respondents in the category rather than by using the smaller number of respondents who could advise against a brand by virtue of their knowledge. If NWOM penetrations are computed only for those consumers who know of poor brands, penetrations will be greater and may then be consistent with the relatively high NWOM frequency figures. The lack of a coherent relationship between frequency and penetration for NWOM may occur because knowledge of negative examples varies widely across categories.

Proposition 4. *The ratio of PWOM to NWOM in a category increases when knowledge of negative examples is limited.*

A priority for further research is the relative *impact* of PWOM and NWOM. If we can measure the impact of PWOM and NWOM on brand choice, we would be able to combine the evidence from this research with findings on incidence and establish a metric for WOM influence. With such a metric, it might be possible to indicate the impact of current WOM on future brand sales.

Proposition 5. *A measure combining the relative incidence and impact of PWOM and NWOM about brands could be an indicator of future changes in brand sales.*

There is evidence in psychological research on judgment that negative information often has a greater impact than positive information. The most common explanation for this effect is that most information is positive, so negative information is rarer and more extreme compared with positive information (Fiske, 1980). The more common positive information may reflect a situation assumed by a recipient, whereas rarer negative information alerts people to facts they do not know.

However, psychological research concerns the formation of judgments, whereas we are interested in brand choice. Evidence in this area is meager, but Ahluwalia (2002) has shown that positive statements about brands have much the same impact as negative statements when the brands are familiar, as in our work. Thus, there is no clear evidence about the relative impact of PWOM and NWOM, and we opt for the following:

Proposition 6. *The impact of PWOM on brand choice is much the same as the impact of NWOM in familiar categories.*

5.4. Limitations

We noted previously a number of problems associated with the measurement of WOM. These problems are difficult to overcome. Our survey-based methods could be prone to biased recall, so it would be helpful to develop other methods to measure WOM to test our findings further.

Our purpose was to establish the main patterns in the reported occurrence of WOM in widely used categories. We excluded rarely used categories and non-users of categories;

also, we did not report on characteristics of WOM, such as whether it occurred between friends or acquaintances, whether it was requested or volunteered, or how strongly it was expressed. More studies are needed on these issues, as well as on the propositions detailed previously.

5.5. Some implications for practice: counteracting NWOM and promoting PWOM

How much effort should practitioners put into combating NWOM? We hesitate to suggest that practitioners should ignore NWOM, despite our evidence that NWOM is less common than PWOM. First, in some categories, NWOM is nearly as common as PWOM, and certain brands (especially those with small market share) may be the object of more NWOM than PWOM. It appears that, sometimes, the failures of a brand can cause NWOM to "take off" and be passed among non-users of the brand. Managers should be worried about this possibility. Second, we find that relatively few people produce NWOM but they do so quite often, which means that marketing efforts to reduce NWOM can focus on a limited number of persons, if these persons can be identified. Third, most of those who produce NWOM are also active producers of PWOM, so these people represent a double hazard. When consumers advise against a brand and recommend a competitive brand, the challenge for the manager is to reverse this pattern of advice.

The practical problem therefore becomes identifying those who give, or who are likely to give, NWOM. We suggest that there are three identifiable groups that may produce NWOM. One group comprises those consumers who complain to the supplier, because this activity is likely associated with negative advice to other consumers as well. A second group includes current users, a minority of whom may advise against their brand. For this group, we need to find out why these users advise against their supplier and whether they do so to all consumers or only to certain types of consumers with particular needs. The third and largest group is past users, who may believe that their current brand is superior to the one they used previously. Normally, companies contact past customers in recovery attempts, but even when these past customers cannot be recovered, it may be worth telling them about improvements in the brand that they previously owned if this would stop them from producing NWOM. All three of these groups can be identified, the first through their complaints and the second and third via databases of current and lapsed customers.

PWOM may be promoted via databases of users and also by using advertising. Bayus (1985) has shown that marketing communications may raise the level of WOM. It seems likely that some messages to customers are particularly effective because they contain ideas that are easily passed on; more research is required to identify the characteristics of such ideas.

5.6. Conclusion

Despite their clear importance in consumer choice, patterns of WOM production are poorly supported by empirical research. We have conducted systematic research in this area. Our multi-category studies identify a number of general patterns of interest to managers and academics alike. We have reported and discussed these patterns and used them to develop six propositions to guide future research about the production of WOM at the category, brand, and individual level. We hope that other marketing researchers will join us in examining this important but neglected area.

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